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Interaction Design for Supporting Energy Researchers to Find Relevant Housing Cooperatives and Energy Declarations

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In Sweden, more than 20,000 active housing cooperatives publish energy declarations and annual reports, providing valuable data for energy researchers. However, accessing specific and relevant information from the large and diverse dataset poses significant challenges. This thesis addresses these challenges using a User-Centred Design (UCD) approach to contribute to the field of interaction design for energy researchers. A web-based search tool was designed and developed through four iterative cycles—covering problem definition, design, development, and testing—to enable energy researchers to access energy-related data efficiently. The findings demonstrate that continuous iterations enhanced the tool's efficiency by improving usability and eliminating redundant design. High scores on the System Usability Scale (SUS) confirm that energy researchers are satisfied with the tool and are willing to use it in their daily workflows. Additionally, this thesis discusses the challenges encountered during this study, broader design implications for supporting other users engaged in practices related to housing cooperatives, and how a proactive approach can foster collaboration and boost research productivity.

SAMMANFATTNING

I Sverige publicerar över 20 000 aktiva bostadsrättsföreningar energideklarationer och årsredovisningar, vilket genererar värdefull data för energiforskare. Att extrahera specifik och relevant information ur detta omfattande och heterogena datamaterial innebär dock avsevärda utmaningar. Denna avhandling adresserar dessa utmaningar genom att tillämpa ett användarcentrerat designperspektiv (User-Centred Design, UCD), i syfte att bidra till forskningsfältet interaktionsdesign inom energiforskning. Ett webbaserat sökverktyg har utformats och utvecklats genom fyra iterativa cykler—innefattande problemformulering, design, utveckling och testning—för att möjliggöra effektiv åtkomst till energirelaterad information för energiforskare. Resultaten visar att kontinuerliga iterationer förbättrade verktygets effektivitet genom ökad användbarhet samt eliminering av redundant design. Höga betyg på System Usability Scale (SUS) bekräftar att energiforskare är nöjda med verktyget och att det med stor sannolikhet kommer att integreras i deras dagliga arbetsflöden. Avhandlingen diskuterar även de utmaningar som uppstod under studiens gång, vidare designimplikationer för att stödja andra användare involverade i verksamheter relaterade till bostadsrättsföreningar, samt hur ett proaktivt förhållningssätt kan främja samarbete och stärka forskningsproduktiviteten.

CCS Concepts: • **Human-centered computing** → **Interaction design theory, concepts and paradigms**.

Keywords: Housing cooperatives, energy research, annual reports, energy declarations, search, user-centred design (UCD)

Nyckelord: Bostadsrättsföreningar, energiforskning, årsredovisningar, energideklarationer, sökning, användarcentrerad design (UCD)

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1 INTRODUCTION

Energy has been a fundamental resource throughout human history, driving societal and economic progress. However, with the acceleration of global industrial development and population growth, energy demand has significantly surged in recent years. Consequently, energy use has emerged as a critical area of focus within Human-Computer Interaction (HCI) research.

In Sweden, reducing energy consumption in the residential and service sector holds significant potential for improving energy efficiency, as this sector accounts for 40% of the nation's total energy consumption [3][4]. Residential property ownership in Sweden is distributed among private individuals, municipal companies, and joint-stock companies, with housing cooperatives comprising approximately 22% of the housing stock – rising to 37% in Stockholm, where this study is based [42].

The Housing cooperative, known as 'Bostadsrättsförening' and abbreviated as 'BRF' in Swedish, is a common form of tenant ownership cooperative in Sweden and the rest of Scandinavia. A BRF owns the entire residential property, and individuals who purchase shares gain access to apartments within the property, automatically becoming members of the cooperative [20]. Members elect the board annually and are responsible for managing everyday tenant matters, including energy usage. Heating and water heating constitute the most significant energy expenses, accounting for 65% of the cooperative's energy consumption [4].

This study is part of CIVIS[1] and BRF Energi[2] initiatives, projects funded by the European Union and the Swedish Energy Agency (Energimyndigheten). These initiatives aim to utilise Information and Communications Technology (ICT) to reduce energy consumption in urban areas [19]. Specifically, these initiatives focus on improving energy management in the housing cooperatives, raising awareness about the environmental impact of energy usage, and promoting knowledge sharing among residents. Both

[1]https://cordis.europa.eu/project/id/608774/results
[2]http://www.brfenergi.se/

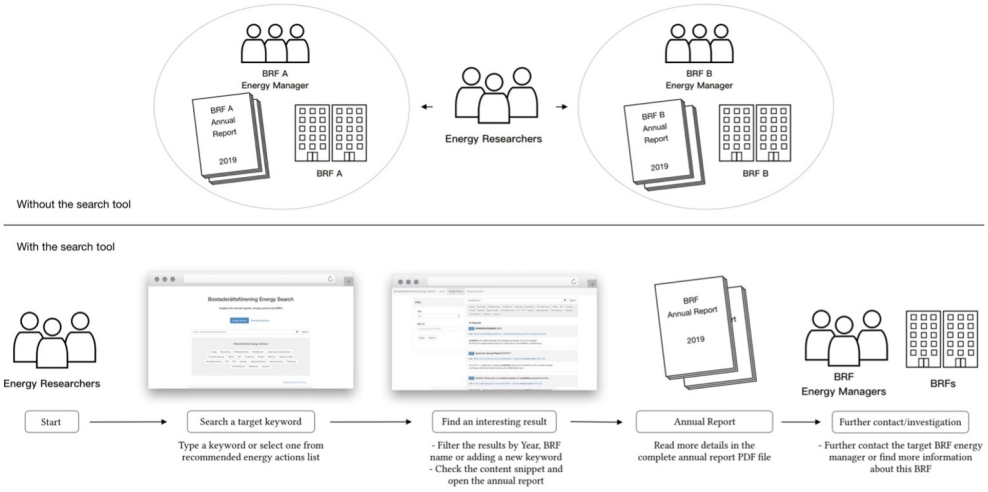


Figure 1: Overview and comparison of energy researchers' workflow with and without the search tool. Energy researchers faced difficulties in extracting information from the thousands of BRFs and their annual reports. The tool enhances the research process by allowing energy researchers to type the keyword, filter the results, and directly navigate to the desired PDF file.

initiatives align with the objectives of Sustainable HCI, addressing the call to "*move beyond the individual*" [7][16]. In recent years, Sustainable HCI research has shifted its focus from targeting individual behaviours to considering the broader ecosystem of stakeholders involved in energy consumption [20]. DiSalvo et al., in 2010, noted that 45% of Sustainable HCI was initially centred on persuasive technology aimed at changing individual behaviour [13]. However, this narrow focus has been criticised [20], leading researchers like Bremer et al. to explore the roles of stakeholders such as housing cooperatives, energy advisors, and policymakers [7].

Many energy-related studies at KTH Royal Institute of Technology focus on stakeholders in energy management in housing cooperatives. Although energy researchers are not direct stakeholders in energy use, their work is critical in shaping decisions about energy-related products, businesses, and policies. However, energy researchers are less explored in previous studies. This study aims to focus on energy researchers and address the gaps by exploring energy researchers' difficulties and identifying how interaction design can support their workflows.

To guide this exploration, the study poses the following research questions:

- **RQ1:** How do researchers interact with housing cooperatives?
- **RQ2:** What challenges do researchers face in their workflows?
- **RQ2:** How can we, as interaction designers, use HCI to support energy researchers?

This thesis seeks to contribute to the area of energy reduction by focusing on energy researchers. This study highlights the importance of effectively acquiring and locating target information from various energy-related documents to support energy researchers' workflows. It details initial design iterations aimed at streamlining researchers' efforts to evaluate the impact of energy reduction actions in housing cooperatives. A newly designed system is introduced, prioritising enhanced access to relevant housing cooperatives and energy-related data. The system has been critically evaluated, with results demonstrating its potential to optimise energy research workflows. Finally, the thesis discusses broader design implications for improving energy research practices. It proposes future design directions to support researchers and other stakeholders, such as energy advisors, in advancing energy reduction initiatives.

2 THEORY AND RELATED RESEARCH

2.1 Researchers and Research Efficiency

This thesis focuses on the primary users of the research – the researchers engaged in multidisciplinary projects such as CIVIS and BRF Energi. In such collaborative projects, researchers from diverse disciplines work together to address a common problem or research question. Collaboration has become a key feature of modern research. Studies have demonstrated that interaction and cooperation between disciplines in the project can enhance understanding and familiarity, ultimately leading to increased research productivity [43] and a more remarkable ability to solve complex scientific problems [41][2]. For example, a study by Lee and Bozeman [26] on universities in the United States and a similar study conducted by Landra [25] on universities in Canada reveals a positive correlation between collaboration and research productivity.

However, collaboration is not without challenges. One of the significant challenges is that researchers from different disciplines often have weak social bonds. Researchers are likely to address the challenge by their methodologies individually and maintain their disciplinary focus [41]. Trust and effective interdependence are hard to develop due to different training backgrounds, advisors, and professional networks [11]. These collaboration silos in research may lead to redundant work and inefficient use of resources.

In a retrospective analysis of an EU-funded multidisciplinary research project [25], the role of HCI researchers was reduced to merely providing visual and evaluation support due to a lack of communication with other disciplines. This resulted in other researchers being unaware of the contributions the HCI domain could offer. To break down the silos in the research collaboration, Bogdan [17] suggests that HCI designers shall "design with researchers". By understanding other disciplines' research contexts and perspectives, processes and tools can be developed and evaluated to serve multidisciplinary projects better. For example, a study on healthcare system research demonstrated that improving processes and tools significantly enhanced the project's quality and efficiency [12].

2.2 HCI Design Approach

Human-computer interaction (HCI) design puts the user at the centre of the stage [34]. It focuses on creating interactive products that support users in their everyday and professional lives [40]. The goal is to bridge the gap between human capabilities and technological complexity, ensuring systems are functional and intuitive. Various HCI design approaches exist, including Activity-Centred Design (ACD) that focuses on the tasks and activities [15]; the predictive method Goal-Directed Design (GDD) [10], the co-creating method Participatory Design (PD) [31], and User-Centred Design (UCD) that prioritises the needs and preferences of the user [24]. While ACD and GDD focus on tasks and objectives, they often overlook user experience [45]. PD, though collaborative, may be impractical when users have limited availabilities. UCD strikes a balance, involving users in design and evaluation without overburdening them, ensuring the final product aligns with users' needs, behaviours, and contexts [34]. Rubin outlines the principles of User-Centered Design (UCD) as an early focus on users and tasks, empirical measurement and testing of product usage, and iterative design enhancements [37]. According to ISO 9241-210, the UCD process involves four phases [1]:

- Understanding the context of use.
- Specifying user needs.
- Producing design solutions.
- Evaluating the design.

To understand the context of use, qualitative research methods such as 1v1 in-depth interviews and focus groups with several people are widely employed [21]. Milena [30] notes that while both the 1v1 interview and focus group methods yield similar results, in-depth interviews often uncover more sensitive topics. To explore "why" rather than "how many" in either process, semi-structured interviews are the most frequently used method in such qualitative research [23]. One of the main advantages is that a semi-structured interview allows the interviewer to dive into a topic through follow-up questions based on the participant´s responses [32]. These interviews effectively combine clarifying questions (e.g., *"Do you mean…?"*) with probing questions (e.g., *"Why?"* or *"How?"*) to gather deeper insights [6].

To continuously gather user input for improving product usability, Beaudouin-Lafon introduces the prototypes, which are critical components of the design process [29]. As *concrete representations* of interactive systems, prototypes help identify functional requirements, usability issues, and performance problems early on. Testing prototypes uncovers design strengths and weaknesses, providing valuable insights for iterative refinement. Throughout the development phase, prototype fidelity evolves from low fidelity (Lo-Fi) to high fidelity (Hi-Fi), reflecting the level of detail and similarity to the final product. Low-fidelity (Lo-Fi) prototypes are simple sketches or mock-ups that emphasise structure and user flow. They allow designers to make significant changes with minimal effort, especially to redirect the design in

the early phase. Conversely, high-fidelity (Hi-Fi) prototypes closely resemble the final product, incorporating detailed interactions and visual elements. However, they require more time and effort to develop. This iterative prototyping process enables designers to refine the product effectively, ensuring it meets user needs while minimising wasted resources.

Iterative prototyping is complemented by continuous usability evaluation to ensure the product meets users' needs. ISO 9241-11 [1] defines Usability as the "extent to which a product can be used by *specified* users, to achieve *specified* goals, with effectiveness, efficiency, and satisfaction in a *specified* context of use." Engaging users throughout the design process to gather feedback and validate design decisions is essential for creating a successful product [33]. Over the past decades, various evaluation methods have been developed. Qualitative methods (e.g., interviews and focus groups) explore users' underlying reasons, motivations, and behaviours. Quantitative methods (e.g., surveys and product analytics) collect and analyse numerical data for measurable insights [14]. Recent research suggests combining these methods provides complementary benefits rather than competition [32][27]. Quantitative methods excel at delivering descriptive data, while qualitative research addresses interpretation challenges [8]. The System Usability Scale (SUS) has been developed as a quick and cost-effective tool to evaluate user interface usability, addressing the typically high costs of quantitative usability studies [9]. SUS consists of a standardised set of questions to assess subjective aspects such as intuitiveness, ease of use, effectiveness, and user satisfaction [8]. Despite focusing solely on users' subjective views, SUS has been proven "valid and reliable" [9] and became an industry standard under ISO 9241-11 in 1998 [1].

2.3 Energy-related Data

Researchers who work with energy consumption often face challenges in accessing relevant data. In the Swedish residential sector, Samuelsson [39] highlights the difficulties energy advisors face when accessing energy-related content for certain housing cooperatives. Energy advisors providing free energy advice to housing cooperatives rely on annual reports and energy declarations to assess energy use before on-site visits. Key energy data in annual reports include yearly energy costs and investments focusing on retrofitting measures [18]. Hasselqvist et al. [24] similarly note the lack of accessible energy data for individuals outside the board of housing cooperatives.

Retrofitting is among the most effective strategies for reducing energy consumption in residential buildings [5]. The Advanced Energy Retrofit Guide (AERG) by the U.S. Department of Energy outlines three approaches, including shallow and deep retrofits:

- Shallow retrofits focus on improvements like envelope insulation, lighting, and heating control. They can achieve energy savings of 25 to 45% with minimal risk and cost [28].
- Deep retrofits: Include measures like advanced building insulation and thermal or water system upgrades, resulting in energy savings of over 45% [22].

By applying retrofitting measures, a residential building in Hammarby Sjöstad in Sweden reduced energy consumption by 50 percent [46]. Retrofitting is typically used at an individual building level, and large-scale retrofitting methodologies are underdeveloped due to different factors such as building age, material, etc [5]. This study developed a data-driven method to optimise energy retrofit decisions for residential buildings at an urban scale. A retrofit knowledge base with consumption data was built to store recommendations for different building types by extracting relevant information for urban scale retrofit modelling. However, this data-driven approach highly depended on the data, the availability of which remained a significant challenge. Pfenninger et al. [35] studied the barriers to open energy data

and identified ethical and security concerns, such as commercial and personal data sensitivity. This study argues that making energy data more accessible could boost research and improve decision-making.

3 METHODS

A user-centric design (UCD) process (See Figure 2) was employed to address the research questions. This process was specifically tailored to address energy researchers' pain points through four iterative improvement cycles. A UX expert evaluated the product in the final iteration and provided detailed feedback for final testing.

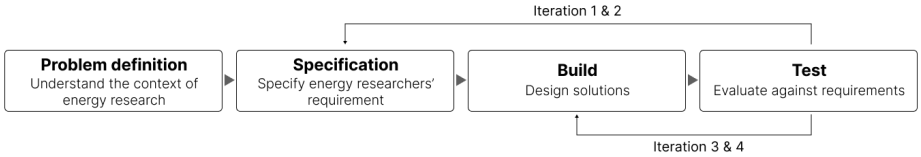


Figure 2: This thesis employs the User-Centered Design (UCD) process. It started with understanding context that energy researchers face in. Iterative development and refinement aim to deliver a product that meets the needs with usability and satisfaction.

3.1 Participants

The project involved four energy researchers (RA, RB, RC and RD) and a UX design expert. RA and RB are actively engaged in CIVIS and BRF Energi initiatives. RC and RD are working with other energy consumption-related research projects and working closely with housing cooperatives. Their participation was distributed across different phases of the process, as shown below (Table 1):

Iteration	Participants	Duration	Setup
	RA	45 mins	Exploratory interview
1	RB	1 hour	Test (lo-fi); Interview
2	RC	1 hour	Test (hi-fi); Interview
3	RA, RD	1 hour	Test (hi-fi); Interview
4	UX Expert	30 mins	Evaluation; Interview

Table 1: Participants who contributed to this thesis projects

3.2 Problem Definition

To understand energy researchers' workflows, mainly how they analyse the energy situation of housing cooperatives, I conducted an exploratory, open-ended interview with an Energy Researcher RA, who is actively engaged in the CIVIS and BRF Energi projects.

To gain deeper insights, after a 20-minute semi-structured interview, I asked the participant to analyse the energy situation of a housing cooperative. Based on these observations, I followed up with a focused semi-structured interview.

3.3 Specification

Based on the notes gathered during the observation and interview, I identified potential design opportunities to support energy researchers' workflows. I sketched several conceptual alternatives to address the identified challenges and prioritised promising and distinct features from the existing tools. This would be the basis for the first conceptualisation of an interactive housing cooperative energy data-focused search engine tailored to researchers' needs.

3.4 Build Phase

As the initial concept, I built a Lo-Fi prototype—a quick and straightforward sketch of user interface scenes and flow. I used Sketch[3] for design and InVision[4] for interaction to make this prototype visual and interactive. The interactive prototype would allow users to click designated buttons to view the corresponding pre-designed outcomes.

3.5 Test Phase

To assess the prototype's ability to meet expectations, I invited Energy Research RB to test the lo-fi prototype. The test was task-oriented: locating information about the energy action "*renovering*" implemented by the housing cooperative "*Brf Masthugget*" in 2014. During this session:

- The total completion time for each step was measured.
- Participants completed a 10-question System Usability Scale (SUS) questionnaire on paper.
- Interaction with the prototype was observed and noted.
- A semi-structured interview was conducted to gather additional insights on usability and concerns.

The entire test lasted approximately 45 minutes, and with the participant's consent, the session was closely observed and audio-recorded for later analysis. Later, Energy Researcher RC in iteration 2 and Energy Researcher RD + RA in iteration 3 conducted the same test but with a different product maturity.

3.6 Iterations

Feedback collected from each test phase informed subsequent design iterations. Depending on the insights, the process cycled back to earlier stages (e.g., Specification or Build) for refinement. After Iteration 1, I revisited the Specification phase to define the new features and developed the prototype using HTML, Python and JavaScript. In Iteration 2, this prototype was tested by Energy Researcher RC, and adjustments were made based on their feedback. For Iteration 3, I invited RA and RD to perform a usability test. In the third iteration, positive feedback from a dual-participant session with RA and RD enabled me to skip the Specification phase in the following iteration and focus directly on addressing interface issues. The semi-structured interviews evolved across the test phase in each iteration to elicit more effective responses.

[3] https://www.sketch.com/
[4] https://www.invisionapp.com/

3.7 Data Analysis

To analyse the data this study has collected, I categorised the data as quantitative or qualitative and applied different analysis methods. Qualitative data was collected through exploratory interviews, usability testing observations, and feedback sessions across the four design iterations. The thematic analysis method was employed systematically to analyse this type of data. Thematic analysis is a type of qualitative analysis used to classify data and identify themes or patterns that emerge from it [22]:

- Familiarisation: Audio recordings of interviews and observation notes were transcribed and reviewed repeatedly to gain an in-depth understanding.
- Coding: Key phrases and insights were labelled with descriptive codes, such as "difficulty locating documents" and "inefficiencies in navigation."
- Theme Development: Related codes were grouped into broader themes, such as "access challenges," "search inefficiencies," and "tool usability improvements."
- Review and Refinement: Themes were reviewed and iteratively refined for clarity and distinctiveness, ensuring the participants accurately represented user experiences.
- Integration: Themes were integrated into the iterative design process, directly informing feature prioritisation and tool refinements.

Additionally, qualitative feedback was used to refine specific features, such as keyword suggestions and snippet previews, ensuring that user needs were addressed comprehensively.

Quantitative data was collected through iterative test phases, task completion time, and SUS scores were noted. To analyse the quantitative data:

- Performance Metrics: Average task completion times were calculated and compared across iterations, highlighting significant reductions in time required for key tasks. To ensure accuracy, adjustments were made to exclude variables outside the scope of the study (e.g., uncontrollable external factors).
- Usability Scores: SUS scores were aggregated and compared to identify improvements in user satisfaction. The progression from acceptable to excellent scores demonstrated the tool's usability enhancements.
- Trend Analysis: Data was visualised to track improvements, such as the percentage of the reduction in task completion time and the change in SUS scores in each iteration.

3.8 Final product evaluation

As the product matured after four iterations, a UX expert was invited to evaluate it remotely due to geographical constraints. This evaluation process included an in-depth review of the system's interface, technical performance, and interaction flow, with feedback categorised into three key areas: System Bugs, UX Writing, and UI Design. Following the evaluation, I implemented improvements to address identified issues, such as fixing bugs, enhancing graphical design, and refining wording for clarity and professionalism. These final refinements were made before the product was delivered.

4 RESULTS

This section will present findings from the interviews, design phase, and UX expert evaluations, contributing to further discussion of the research questions posed.

4.1 Interview

4.1.1 How Energy Researchers Study Housing Cooperative. Energy researchers studying housing cooperatives are particularly interested in what energy reduction actions have been taken, how much these actions cost, and what subsequent effects have been had. While direct visits and interactions with the boards provide valuable insights, researchers often gather information beforehand by reviewing publicly available resources: annual reports and energy declarations. These documents are commonly referenced to facilitate discussions with energy managers.

> ”I usually start with the annual report.”

Energy Researcher RA emphasised the importance of annual reports, describing them as “*the doors to start analysing relevant energy action cases for housing cooperatives*.” In Sweden, housing cooperatives are legally required to publish annual reports, which include energy consumption and cost data. These documents offer extensive insights into energy practices and efficiency. RA further noted,

> ”These annual reports can briefly help us find good ideas about what has been implemented and the effects.”

Additionally, energy declarations, which are mandatory for all buildings in Sweden since 2009 and updated every ten years, offer *”great help”* in analysing the energy performance of housing cooperatives.

> “But these are not enough for deeper studies”, and “we need to contact the energy manager and visit on-site.”

4.1.2 Challenges in Finding Information on the Web and Locating in the Documents. RA described locating specific housing cooperatives and their published documents online as “*challenging*”. With 26,000 active housing cooperatives in Sweden publishing reports annually, *“thousands of annual reports*” are available. Further complicating the process, *“many BRFs have similar names*”, making searches prone to errors.

With the correct annual reports from the certain BRF, *“Reading those poor qualities, 30-page long annual reports can sometimes be painful”*. RA further notes, *“Energy-related information is usually just a small part of these long reports”* and *“poor-quality scanning makes many documents unreadable.”*

“You can’t search” The lack of keyword search for poorly scanned PDFs forces researchers to manually skim through reports, which is time-consuming and inefficient. This challenge initially described by RA is reiterated by subsequent participants RB, RC and RD as *“annoying”* and *“painful”*.

4.1.3 Lacking ICT Skills and Support. “I heard about a technique” Energy Researcher RB referred to as Optical Character Recognition (abbreviated as OCR), which emerged as a promising technology for converting poorly scanned PDFs into searchable data. RB reported failing to find good OCR software capable of efficiently processing and searching poorly scanned PDFs. Then, he sought to develop a technical solution to process these poorly scanned PDF files and make these files searchable but eventually “*abandoned the idea”* due to a lack of software development knowledge.

> “I am not a coding professional”, “it was not easy to learn.”

RB noted, *“I didn’t ask for help because it felt too troublesome.”* This highlights a broader issue: researchers may avoid seeking assistance from software developers or HCI designers due to disciplinary silos and perceived barriers.

	Local OCR solution	**Online solution with Google custom API**
Files storage	Local storage of PDF files in the application	No local storage needed; files remain in original online locations
Files updating	Manual or scripted updates required; time consuming	Real-time searching by Google; files are always considered as up-to-date
OCR processing	Required; time-intensive and demands high performance	Depends on Google; processing time varies
Index establishing	Necessary	Not feasible
Sustainability	Ongoing maintenance required	Minimal maintains needed

Table 2: Participants who contributed to this thesis projects

4.2 Design Opportunity and Solutions

The process of finding information for energy researchers was described as "*vicious*", "*tedious* and "*discouraging*", requiring time spent on reading long, poor-quality documents, plus both expertise in housing cooperative documents and facility with energy-related terminology. This presented an opportunity for interaction designers to address inefficiencies and improve the workflow.

4.2.1 Proposed Solution: A Thematic Search Engine. The primary challenge identified was the inefficient access to energy-related data across thousands of online documents. To address this, I designed a search engine that would locate energy-related keywords in annual reports, offer filters to narrow the search results, and provide snippets of content to give users context before opening full documents.

4.2.2 Optimising PDF Search with OCR. I recognised the concern about the searching ability in those poorly scanned PDFs and explored technical solutions to tackle the issue. Optical Character Recognition (abbreviated as OCR) emerged as a promising technology for converting poorly scanned PDFs into searchable data. Two implementation paths were considered (See Table 2):

- Local OCR solution: Establishes an index for navigating specific results.
- Online API: This option offers broader coverage, higher accuracy, and faster response times while eliminating the need for database maintenance.

Based on these considerations, I evaluated several online OCR solutions available at the time, including ABBYY FineReader, Adobe Acrobat DC OCR, Tesseract OCR and Google Cloud Vision API. I selected a Google-powered solution as it offered the best balance of cost, recognition accuracy and ease of integration. This solution was then integrated into the application to provide reliable text recognition and efficient search functionality.

4.2.3 Features Added: Housing Cooperative Selector and Pre-defined Energy-related Keywords. During the iterations, two additional features were added. In the 1st iteration, RB accidentally spelled the name of the BRF. This caught my attention; RB explained that *"many BRFs have similar names"* nationwide. To avoid incorrect spelling or a housing cooperative with a *"similar but wrong name"*, I integrated a selector in the filter containing 9,000 active housing cooperatives sourced using a web crawler built with Python. This selector includes autocomplete functionality and provides suggestions based on user input, reducing errors caused by similar names or types. In the following iteration, Energy Researcher RC appreciated the selector and described it as *"great support"*.

In the 2nd iteration, RC identified an interesting pattern: *"certain energy actions were frequently investigated within the context of BRFs"*. Based on these insights, I added a pre-defined list of the 20 most common retrofitting keywords. With these keywords, users can select from this list instead of manually typing, and the list is stored in an editable file for future updates. In the 3rd iteration, RA thought this feature was *"a cool idea"*; RD further described it as *"helpful hints"* that inspired users.

4.3 Usability Improvements Across Iterations

4.3.1 Efficiency Gains. During the Test phase in each iteration, the completion times for identical tasks were conducted (See Figure 3). Without technical tools, using RA's regular search method, Google to locate the energy keyword *"Renovering"* in the 2014 annual report for *"Brf Masthugget"* took 82.4 seconds, 60,1 seconds of which was spent on finding the correct yearly report.

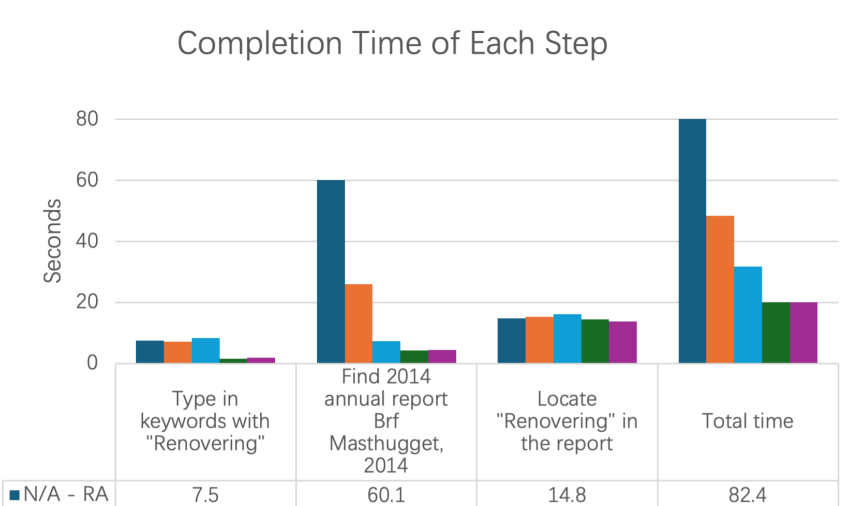


Figure 3: Measurement of completion time with four participants in three iterations.

With keyword suggestions integrated, the time spent on the first step, *"Type in keywords with 'Renovering'"*, drops from 7.1 and 8.3 to 1.5 and 1.8 seconds. This leads to an approximately 80% reduction for one step.

For the second step, *"Find 2014 annual report for 'Brf Masthugget'"*, RA spent 60,1 seconds locating the correct annual report by Google, while RB mistyped the name, 25.9 seconds is still a significant improvement. After adding an autocomplete housing cooperative selector and a year selector, RC, RD, and RA quickly located *"Brf Masthugget"* with 7.3, 4.2 and 4.5 seconds. The time saving for this step is about 93%. For the third step, *"Find 'Renovering' in the report"*, each participant spent a similar amount of time completing this task.

Overall, the task completion time dropped from 82.4 seconds with Google to 20 seconds with the final product; RA experienced a 75% reduction. Since we developed the lo-fi prototype in the first iteration,

the completion time dropped from 48.3 seconds with the lo-fi prototype to 20 seconds and 20.1 seconds. This 55% reduction demonstrates significant efficiency improvements through iterative design.

4.3.2 Satisfaction improvement . During the test phase in each iteration, the SUS, consisting of 10 standard questions, was answered, and the scores were calculated (See Figure 4).

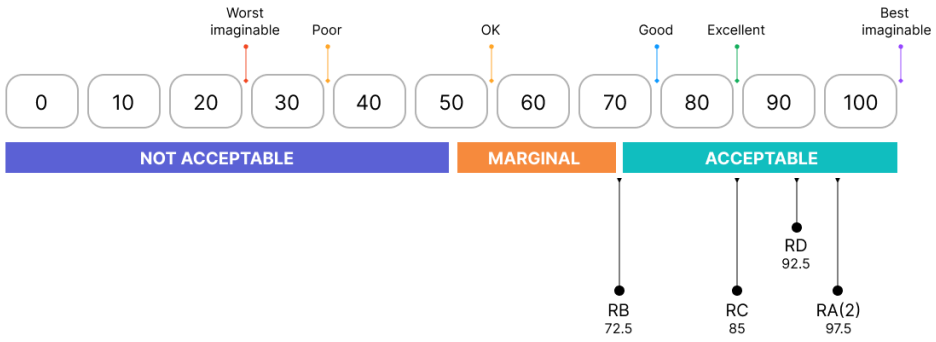


Figure 4: The System Usability Scale (SUS) shows an "Excellent" standard on the usability

According to ISO 9241-11 [1], scores above 70 are considered "Acceptable", 75 as "Good", 85 as "Excellent", and 100 as "best imaginable". All participants scored who scored above 71 on the System Usability Scale (SUS), surpassing the line for "Acceptable" usability. Scores increased from 72.5 in early iterations to 92.55 and 90 in the final iteration, reflecting "excellent" usability.

"I think this can be a helpful concept" – RB

Participants prised:

- Keyword suggestions. The suggested keywords covered standard energy terms in annual reports and streamlined searches.
- Filters and snippets: Simplified finding specific information and provided helpful context.
- Interface design. RA and RC highlighted the product's clear, straightforward presentation and clean, visually appealing interface.

RA stated, *"I believe this tool will definitely improve our work with housing cooperatives when we need to find information."* RA further summarised the sentiment by saying, *"This will make our lives easier."*

4.4 Final Product Evaluation

The final product evaluation was conducted with a UX design expert to assess its usability, functionality, and overall user experience. The purpose was to identify remaining areas for improvement and ensure the product's readiness for practical deployment.

The evaluation highlighted several technical issues that needed resolution to ensure a seamless user experience. For example, the year selector occasionally failed to register user input; the snippet occasionally displayed incorrect or garbled text, notably the Swedish alphabet "Å, Ä, Ö". The expert emphasised the importance of clear and concise UX writing to ensure users can interpret the system intuitively. Key areas for improvement included the wording accuracy, misleading slogans and spelling errors. The evaluation also revealed several UI-related issues impacting accessibility, usability and aesthetics, such as

the inconsistent font size, colour intensity of snippets and loading indicators. The text size on the result page was considered too small for comfortable reading. The expert praised that the attempt could successfully address the pain points of energy researchers by reducing the time, simplifying the process, providing a straightforward interface, and highlighting its usability and problem-solving potential. The expert also recommended potential enhancements for future iterations, such as optimisation for mobile devices, which can ensure full compatibility to expand accessibility for users working in the field.

Based on the feedback, further refinements were implemented (see Figure 5). After that, the product was ready for deployment, and it had the potential to evolve further based on real-world usage and feedback.

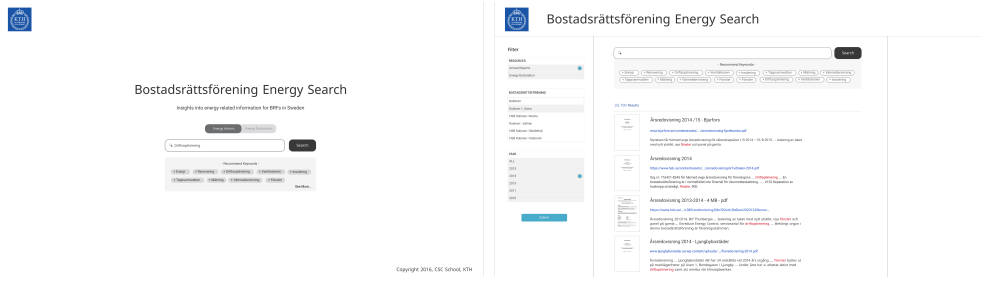


Figure 5: Screenshots of the final product. Left: Homepage where user type in the keywords or select multiple keywords from the suggestion box. Right: Result page with filters on the left side where user can select multiple filters

5 DISCUSSION

In this study, we have explored how energy researchers at KTH Royal Institute of Technology engage with housing cooperatives, their challenges during the workflow, and the potential of a digital tool to streamline their research workflows. By employing a User-Centric Design process, we addressed specific pain points related to accessing and analysing energy-related documents such as annual reports and declarations. The findings highlight the proposed solution's successes and limitations and its broader applicability and potential for future improvement.

5.1 Broader Applicability Beyond the Study

This study primarily focuses on researchers working with housing cooperatives in Sweden. One could argue that the focused user group is narrow and not representative of other users, and this study cannot be expanded to other areas. Similar challenges in accessing annual reports and energy declarations are reported by other users, such as independent advisors [20] and municipal energy advisors [39]. These professionals rely on such documents to assess energy situations and advise energy managers on reduction strategies at the housing cooperatives. With this proposed tool, energy advisors could skip the tedious ordering processes [39] and directly locate the information they need, enabling them to focus on the quality and effectiveness of their energy advice. This efficiency could boost engagement and support for energy reduction initiatives [20]. We believe the tool's applicability can be extended. This tool could be adapted for broader applications with minor modifications to the code.

Beyond the energy domain, the annual report interests potential apartment buyers. In Sweden, people who consider purchasing an apartment have an interest in evaluating the financial situation of the housing

cooperative, such as the debt that impacts the common monthly charge and the energy-related investment to assess if or not there will be retrofitting that will increase the monthly cost.

5.2 Supporting Multidisciplinary Research

One of the key insights from this study is the role of collaboration in multidisciplinary research, particularly between energy researchers and interaction designers. Multidisciplinary research thrives on knowledge exchange, innovation, and resource sharing, enabling complex problems to be tackled from diverse perspectives [44]. However, a notable observation from this study is the hesitation among researchers to seek assistance from other disciplines. For instance, one researcher attempted to address technical challenges independently rather than collaborating with HCI designers or software engineers within the project.

This highlights a border issue: disciplinary silos and a lack of mutual understanding in daily practices. This aligns with a study showing that it often manifests only at the final stages of research, where deep integration across disciplines produces meaningful societal impact [38]. In daily practice, there remains a lack of mutual understanding and attention among researchers from different fields [11]. An interaction designer is trained to understand the user and design a product that can solve the user's needs and meet the user's expectations. With being proactive in supporting other researchers, the help from HCI professionals can, therefore, be expected to boost the productivity of the research. Other researchers gain a deeper understanding of HCI and its value through design with the user and design for the user. They can, therefore, realise the need for assistance or collaboration in future projects.

5.3 Challenges and Limitations

One persistent challenge was improving the time required to locate specific document content. Despite efforts to streamline access, the "*Locate 'Renovering' in the report*" step shows limited improvement. While the initial design planned to have a feature that would highlight the exact section containing the relevant content and help the user open with the target content in the length document, the chosen OCR solution, Google API, does not support this advanced indexing functionality, as well as data extraction, and semantic processing. Moving forward, future work could explore more advanced approaches, such as natural language processing (NLP) and machine learning., which are capable of processing, structuring and summarising key information. We believe incorporating such intelligent features could significantly reduce the time users spend reading documents and improve efficiency further.

Another challenge is the reliance on annual reports and energy declarations. The annual reports are often criticised for being indirect and complicated to interpret [24], the energy declarations are often criticised for being infrequently updated since they are revised only every ten years" [24], and their quality varies significantly [39]. Furthermore, many energy-related actions require years before their effects are visible, limiting the immediate utility of these documents. Data privacy concerns and the lack of mandatory publication requirements constrain access to more granular, real-time, or higher-quality data. To address this limitation, future research could advocate for policy changes encouraging the publishing of more detailed and standardised energy data or explore partnerships with housing cooperatives to collect additional data voluntarily.

A further limitation was the delayed progress in development, which led to the tool not being published on time. As a result, it missed a critical timing window when energy researchers at CIVIS and BRFEnergi projects had a strong need for navigating energy related data from large volumes of housing

cooperative. This timing mismatch reduced the opportunity for real-world validation. Future iterations should prioritise aligning development with user needs to ensure timely deployment and maximise the tool's timing relevance.

5.4 Method Retrospective

This study successfully addressed a pressing pain point through an iterative design process and improved efficiency and satisfaction scores. The User-Centred Design (UCD) was performed as the primary methodology. During the UCD process, exploratory semi-structured interviews were conducted to understand the user context and the feedback of the prototype, and thematic analysis was applied to analyse the collected qualitative data; task completion time and System Usability Scale (SUS) were used to analyse the quantitative data. These methodologies successfully prove their ability to progress the study in this study. The methods can be expanded to gain broader applicability and support multidisciplinary research.

However, this study's exploratory phase relied on a single interview with one researcher, which poses risks in HCI design. Although the pain points identified were later confirmed by other participants, relying on a limited sample size could lead to too narrowly tailored solutions. Ideally, the exploratory phase should have included interviews or a focus group with several researchers to provide a more comprehensive understanding of user needs [36]. Additionally, the evaluation phase could have benefited from involving more energy researchers for a broader assessment. Expanding the participant pool in future studies would enhance the robustness of the findings and reduce the risk of bias.

5.5 Future Research

One key area is expanding the application of HCI to address additional challenges researchers face throughout their workflows. To achieve that objective, future studies should observe more research activities and interview more participants to gain broader and deeper insights into their workflows and challenges. Beyond energy researchers, including other energy-related stakeholders such as municipal energy advisors and energy managers in housing cooperatives in interview and usability testing could provide valuable exploratory perspectives, supporting the adaptation of the tool for a wider audience and enhancing its societal relevance. Additionally, further exploration is needed to integrate advanced technologies such as natural language processing (NLP) and machine learning into tools that support predictive modelling, automated data extraction, constructing reliable summarises and enhanced visualisation, improving efficiency and usability. Another area of interest is applying HCI methods to foster collaboration across disciplines and lead to innovative solutions.

6 CONCLUSIONS

This thesis has investigated how energy researchers engage with housing cooperatives, their challenges, and the potential for a digital tool to streamline their workflows. By employing a User-Centric Design (UCD) approach, key pain points have been identified and addressed through the iterative development of a web-based search tool tailored to energy researchers' needs. This study reveals significant challenges, including difficulty locating specific reports online, navigating lengthy and poorly scanned documents, and limited access to accurate energy data. Through iterative design, the tool improved task efficiency by up to 75% and achieved excellent usability scores on the System Usability Scale (SUS). Features such

as OCR-enabled search, keyword suggestions, autocomplete filters, and snippet previews significantly enhanced usability and reduced manual effort.

Beyond energy researchers, we believe the tool has broader applicability for energy advisors, policymakers, and prospective apartment buyers, addressing similar needs for accessing housing cooperative data. Minor adaptations to the tool could expand its functionality for these user groups. The thesis also highlights the importance of collaboration in multidisciplinary research. Interaction designers can be crucial in addressing research challenges, fostering productivity, and creating tools that facilitate collaboration and innovation.

However, challenges remain, such as improving the accuracy of locating specific data in documents and addressing the limitations of available energy data. Future work could explore more advanced technical solutions, expand the scope to other stakeholders, and advocate for better data availability through policy changes. This study demonstrates the potential of HCI to address research challenges in energy workflows, providing a foundation for future work in advancing sustainable energy practices and multidisciplinary collaboration through user-centred tools.

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REFERENCES

[1] 1998. Ergonomic requirements for office work with visual display terminals (VDTs) – Part 11: Guidance on usability.

[2] 2004. *Facilitating Interdisciplinary Research*. National Academies Press, Washington, D.C. 978-0-309-09435-1 https://doi.org/10.17226/11153 Pages: 11153.

[3] 2017. *Climate change 2007: Residential and Commercial buildings (Chapter 6)*. Technical Report. IPCC (Intergovernmental Panel on Climate Change).

[4] 2017. *Energiläget 2017*. Technical Report. The Swedish Energy Agency, Eskilstuna.

[5] Usman Ali, Mohammad Haris Shamsi, Mark Bohacek, Cathal Hoare, Karl Purcell, Eleni Mangina, and James O'Donnell. 2020. A data-driven approach to optimize urban scale energy retrofit decisions for residential buildings. *Applied Energy* 267 (June 2020), 114861. https://doi.org/10.1016/j.apenergy.2020.114861

[6] Ahlem Assila, Káthia Marçal De Oliveira, and Houcine Ezzedine. 2014. Towards Qualitative and Quantitative Data Integration Approach for Enhancing HCI Quality Evaluation. In *Human-Computer Interaction. Theories, Methods, and Tools*, David Hutchison, Takeo Kanade, Josef Kittler, Jon M. Kleinberg, Alfred Kobsa, Friedemann Mattern, John C. Mitchell, Moni Naor, Oscar Nierstrasz, C. Pandu Rangan, Bernhard Steffen, Demetri Terzopoulos, Doug Tygar, Gerhard Weikum, and Masaaki Kurosu (Eds.). Vol. 8510. Springer International Publishing, Cham, 469–480. 978-3-319-07232-6 978-3-319-07233-3 https://doi.org/10.1007/978-3-319-07233-3_43 Series Title: Lecture Notes in Computer Science.

[7] Christina Bremer, Bran Knowles, and Adrian Friday. 2022. Have We Taken On Too Much?: A Critical Review of the Sustainable HCI Landscape. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–11. 978-1-4503-9157-3 https://doi.org/10.1145/3491102.3517609

[8] John Brooke. 1995. SUS: A quick and dirty usability scale. *Usability Eval. Ind.* 189 (Nov. 1995), 189–194.

[9] John Brooke. 2013. SUS: a retrospective. *Journal of Usability Studies* 8 (Jan. 2013), 29–40.

[10] Alan Cooper, Robert Reimann, Dave Cronin, and Alan Cooper. 2014. *About face: the essentials of interaction design* (fourth edition ed.). John Wiley and Sons, Indianapolis, IN. 978-1-118-76657-6

[11] Jonathon Cummings and Sara Kiesler. 2008. Collaborative Research across Disciplinary and Organizational Boundaries. In *Scientific Collaboration on the Internet*. Vol. 35. 99–118. https://doi.org/10.1177/03061270505535 Journal Abbreviation: Scientific Collaboration on the Internet.

[12] Denise H. Daudelin, Harry P. Selker, and Laurel K. Leslie. 2015. Applying Process Improvement Methods to Clinical and Translational Research: Conceptual Framework and Case Examples. *Clinical and Translational Science* 8, 6 (Dec. 2015), 779–786. https://doi.org/10.1111/cts.12326

[13] Carl DiSalvo, Phoebe Sengers, and Hrönn Brynjarsdóttir. 2010. Mapping the landscape of sustainable HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. Association for Computing Machinery, New York, NY, USA, 1975–1984. 978-1-60558-929-9 https://doi.org/10.1145/1753326.1753625

[14] L. Eronen. 2001. Combining Quantitative and Qualitative Data in User Research on Digital Television. In *the 1st Panhellenic Conf. with Int. Participation on Human-Computer Interaction, PC-HCI 2001, Patras Greece, Dec. 7-9, 2001*. 51–56.

[15] Bernard R. Gifford and Noel D. Enyedy. 1999. Activity centered design: towards a theoretical framework for CSCL. In *Proceedings of the 1999 conference on Computer support for collaborative learning (CSCL '99)*. International Society of the Learning Sciences, Palo Alto, California, 22–es.

[16] Lon Åke Erni Johannes Hansson, Teresa Cerratto Pargman, and Daniel Sapiens Pargman. 2021. A Decade of Sustainable HCI: Connecting SHCI to the Sustainable Development Goals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–19. 978-1-4503-8096-6 https://doi.org/10.1145/3411764.3445069

[17] Hanna Hasselqvist and Cristian Bogdan. 2014. Design with people in multi-disciplinary, multi-cultural project consortia : A set of Guidelines. In *:*. KTH, Media Technology and Interaction Design, MID. http://www.drs2014.org/en/

[18] Hanna Hasselqvist, Cristian Bogdan, and Filip Kis. 2016. Linking Data to Action: Designing for Amateur Energy Management. In *Proceedings of the 2016 ACM Conference on Designing Interactive Systems*. ACM, Brisbane QLD Australia, 473–483. 978-1-4503-4031-1 https://doi.org/10.1145/2901790.2901837

[19] Hanna Hasselqvist, Cristian Bogdan, Mario Romero, and Omar Shafqat. 2015. Supporting Energy Management as a Cooperative Amateur Activity. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. ACM, Seoul Republic of Korea, 1483–1488. 978-1-4503-3146-3 https://doi.org/10.1145/2702613.2732724

[20] Hanna Hasselqvist and Elina Eriksson. 2018. Designing for diverse stakeholder engagement in resource-intensive practices. In *Proceedings of the 10th Nordic Conference on Human-Computer Interaction*. ACM, Oslo Norway, 426–438. 978-1-4503-6437-9 https://doi.org/10.1145/3240167.3240193

[21] Immy. Holloway. 2005. *Qualitative research in health care*. Open University Press, Maidenhead, England ;. 978-0-335-21294-1

[22] Ljubomir Jankovic. 2019. Lessons learnt from design, off-site construction and performance analysis of deep energy retrofit of residential buildings. *Energy and Buildings* 186 (March 2019), 319–338. https://doi.org/10.1016/j.enbuild.2019.01.011

[23] Hanna Kallio, Anna-Maija Pietilä, Martin Johnson, and Mari Kangasniemi. 2016. Systematic methodological review: developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing* 72, 12 (Dec. 2016), 2954–2965. https://doi.org/10.1111/jan.13031

[24] Eeva Kangas and Timo Kinnunen. 2005. Applying user-centered design to mobile application development. *Commun. ACM* 48, 7 (July 2005), 55–59. https://doi.org/10.1145/1070838.1070866

[25] R jean Landry, Namatie Traore, and Beno t Godin. 1996. An econometric analysis of the effect of collaboration on academic research productivity. *Higher Education* 32, 3 (Oct. 1996), 283–301. https://doi.org/10.1007/BF00138868

[26] Sooho Lee and Barry Bozeman. 2005. The Impact of Research Collaboration on Scientific Productivity. *Social Studies of Science* 35 (Oct. 2005), 673–702. https://doi.org/10.1177/0306312705052359

[27] Gitte Lindgaard. 2006. Notions of thoroughness, efficiency, and validity: Are they valid in HCI practice? *International Journal of Industrial Ergonomics* 36, 12 (Dec. 2006), 1069–1074. https://doi.org/10.1016/j.ergon.2006.09.007

[28] Guopeng Liu, Bing Liu, Weimin Wang, Jian Zhang, Rahul A. Athalye, Dave Moser, Eliot Crowe, Nick Bengtson, Mark Effinger, Lia Webster, and Mike Hatten. 2011. *Advanced Energy Retrofit Guide Office Buildings*. Technical Report PNNL-20761, 1028081. PNNL–20761, 1028081 pages. https://doi.org/10.2172/1028081

[29] Wendy Mackay and Michel Beaudouin-Lafon. 2009. Prototyping Tools and Techniques. In *Human-Computer Interaction*, Andrew Sears and Julie Jacko (Eds.). Vol. 20094116. CRC Press, 121–143. 978-1-4200-8890-8 978-1-4200-8889-2 https://doi.org/10.1201/9781420088892.ch7 Series Title: Human Factors and Ergonomics.

[30] Zaharia Milena, Grundey Dainora, and Alin Stancu. 2008. Qualitative research methods: A comparison between focus group and in-depth interview. *Annals of Faculty of Economics* 4 (May 2008), 1279–1283.

[31] Michael Muller and Allison Druin. 2002. Participatory Design: The Third Space in HCI. *Handbook of HCI* (Jan. 2002).

[32] World Health Organization. 2004. How to investigate the use of medicines by consumers / Anita Hardon, Catherine Hodgkin, and Daphne Fresle. Pages: WHO/EDM/PAR/2004.2.

[33] Derya Ozcelik, Javier Quevedo-Fernandez, Jos Thalen, and Jacques Terken. 2011. Engaging users in the early phases of the design process: attitudes, concerns and challenges from industrial practice. In *Proceedings of the 2011 Conference on Designing Pleasurable Products and Interfaces*. ACM, Milano Italy, 1–8. 978-1-4503-1280-6 https://doi.org/10.1145/2347504.2347519

[34] Roy Pea. 1987. User Centred System Design-New Perspectives on Human/Computer Interaction. *J Educ Comput Res* 3 (Jan. 1987).

[35] Stefan Pfenninger, Joseph DeCarolis, Lion Hirth, Sylvain Quoilin, and Iain Staffell. 2017. The importance of open data and software: Is energy research lagging behind? *Energy Policy* 101 (Feb. 2017), 211–215. https://doi.org/10.1016/j.enpol.2016.11.046

[36] Fatemeh Rabiee. 2004. Focus-group interview and data analysis. *Proceedings of the Nutrition Society* 63, 4 (Nov. 2004), 655–660. https://doi.org/10.1079/PNS2004399

[37] Jeffrey Rubin and Dana Chisnell. 2008. *Handbook of usability testing: how to plan, design, and conduct effective tests* (2nd ed ed.). Wiley Pub, Indianapolis, IN. 978-0-470-18548-3

[38] Meghna Sabharwal. 2013. Comparing Research Productivity Across Disciplines and Career Stages. *Journal of Comparative Policy Analysis: Research and Practice* 15, 2 (April 2013), 141–163. https://doi.org/10.1080/13876988.2013.785149

[39] David Samuelsson. 2018. *Identifying Opportunities for Digital Tools to Support Energy Advisors Working with Housing Cooperatives*. Master's thesis. KTH, Media Technology and Interaction Design, MID. Backup Publisher: KTH, Media Technology and Interaction Design, MID Issue: 2018:394 Series: TRITA-EECS-EX.

[40] Helen Sharp, Yvonne Rogers, and Jennifer Preece. 2007. *Interaction Design. Beyond Human-Computer Interaction*. 978-0-470-01866-8

[41] Diane H. Sonnenwald. 2007. Scientific collaboration. *Annual Review of Information Science and Technology* 41, 1 (Jan. 2007), 634–681. https://doi.org/10.1002/aris.2007.1440410121

[42] Statistics Sweden. 2014. Statistikdatabasen. http://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START__BO__BO0104/BO0104T04/?rxid=d4abd6aa-84eb-41dbb9d8-324c83482584

[43] Daniel Stokols, Shalini Misra, Richard P. Moser, Kara L. Hall, and Brandie K. Taylor. 2008. The Ecology of Team Science. *American Journal of Preventive Medicine* 35, 2 (Aug. 2008), S96–S115. https://doi.org/10.1016/j.amepre.2008.05.003

[44] Caroline S. Wagner. 2018. *The Collaborative Era in Science: Governing the Network*. Springer International Publishing, Cham. 978-3-319-94985-7 978-3-319-94986-4 https://doi.org/10.1007/978-3-319-94986-4

[45] Ashley Williams. 2009. User-centered design, activity-centered design, and goal-directed design: a review of three methods for designing web applications. In *Proceedings of the 27th ACM international conference on Design of communication*. ACM, Bloomington Indiana USA, 1–8. 978-1-60558-559-8 https://doi.org/10.1145/1621995.1621997

[46] Helene Wintzell, Allan Larsson, and Willy Ociansson. 2013. *Under 100 - att lyckas med energi i Hammarby Sjöstad*. Technical Report.

Appendix 1: Interview Plan (17/11/2016)

Procedure:

1. Semi-structured interview plan.
2. Consent form specifying the rules of the interview and the confidentiality commitment.
3. Interviewee: TBD

   Date: 17/11/2016 13:30.

   Duration: approx. 45mins.

   Place: TBD, Stockholm
4. Send the consent form, the interview plan with summary to the interviewee by 2 hours before the interview.
5. Prepare equipment for the interview.

   Demonstration: Laptop,

   Recording: Mobile phone,

   Taking notes: pen and paper.
6. Contact the interviewee again to confirm the date and location of the interview.

Thesis Introduction:

In Sweden, here are over 30,000 registered BRFs. Most energy managers are not professional, or we can say amateurs in energy management and energy decision. In this case, there is a large waste of energy. As for energy experts, they are hard to help a BRF to optimize the energy consumption since it's not easy to find interesting BRFs and interesting energy actions from a large number of BRFs and their annual reports or energy declaration. It always take a long time to do so. So my work is to develop a semi-automatically tool for supporting energy experts find the cases they want from this big amount of annual reports.

So my interview will focus on how you interact with BRFs and energy actions or energy declaration, how you help BRFs optimize the energy consumption, how do you think of the problem of finding specific cases and then how do you understand a tool for supporting energy experts, like you to find out interesting cases.

The Background and Problems

● Could you please tell me anything about your work, how you work with energy actions and BRFs? ● What is current BRFs energy situation? ● What is a BRF's energy action? Why is it interesting for you? ● How do you find out a relative energy action when you are helping a BRF optimize the energy action?	● How did you learn about these problems? ● Why is this considered a problem? ● What is the scope of the problem?	● Can you expand a little on this? ● Can you tell me anything else? ● Can you give me some examples?

● What are you looking for when looking at a BRFs energy situation? ● Did you have concrete cases of energy experts that you met that had an interaction with BRFs? How did they do with BRFs? And what were problems they had? ● What do you think about an annual report? Is it useful for you to find relative energy keywords? ● Can you tell me how existing energy actions, for example listed in annual reports affect following energy actions? ● Can you tell me something about the problems in finding interesting energy actions and BRFs? ● Which problems affect your area of research in interacting with BRFs and finding energy actions?		
● In your experience, which problems bother energy experts, like you, helping BRFs optimizing energy situation the most? ● What would your sources of information be, and whether annual reports among with you? ● In your experience, which problems bother energy experts, like you, finding a data from a large number of annual reports?	● Why? ● How did you solve these problems?	

Usability Test

Open TBD (prototype link)

A brief introduction about the prototype testing

Describe the task step by step:

1. Search for energy actions. Type the keyword of energy action <Renovering>.
2. Show the result list.
3. Find the annual report for a housing cooperative <Brf Masthugget>.
4. Find it in the year <2014>.
5. Show the result list.
6. Open the PDF file.
7. Find the keyword and locate it in the annual report.
8. Show the result.

The conductor starts measuring time and starts observing

The conductor stops measuring time when the last task is finished

There are five minutes for the participant to free use. The conductor observes the entire using process.

● What was your first impression of the webpage? ● Could you find what you were looking for? How did you think when using the site? ● Was there something missing you were expecting to see? Examples? ● Was it easy to get into the tool on each page? ● Did you notice…. (keyword recommendation list, tick box for years, for different BRFs) ? ● What do you think the intended audience is?	● Why? ● Was there something that could have been added to increase the return visitor rate? ● Any other elements of the site of design you need to ask? ● Was everything organized well and set out as expected?	● Can you expand a little on this? ● Can you tell me anything eles? ● Can you give me some examples?
● What would encourage you to return to this site in the future? ● What do you like or dislike about this webpage? ● If you could change one thing on the site, or add a new thing on the site, whether it is major or minor, what would be at the top of the to do list?	● Why? ●	

Appendix 2: System Usability Scale (SUS)

Please enter your participant number: _____

System Usability Scale (SUS)

Please select the answer that best expresses how you feel about each statement after using the website today.

	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
1. I think I would like to use this tool frequently.	☐	☐	☐	☐	☐
2. I found the tool unnecessarily complex.	☐	☐	☐	☐	☐
3. I thought the tool was easy to use.	☐	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5. I found the various functions in this tool were well integrated.	☐	☐	☐	☐	☐
6. I thought there was too much inconsistency in this tool.	☐	☐	☐	☐	☐
7. I would imagine that most people would learn to use this tool very quickly.	☐	☐	☐	☐	☐
8. I found the tool very cumbersome to use.	☐	☐	☐	☐	☐
9. I felt very confident using the tool.	☐	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this tool.	☐	☐	☐	☐	☐

How likely are you to recommend this website to others? (please circle your answer)

Not at all likely 0 1 2 3 4 5 6 7 8 9 10 Extremely likely

Appendix 3: Consent From

Consent form

Master Thesis Project:

Interaction Design for Supporting Energy Experts to Find Relevant Housing Cooperatives

Project context

In order to support energy experts to help housing cooperatives make appropriate energy decisions which will optimize and reduce the energy consumption in their cooperative, this project aims to use interaction design methods to understand users, the energy experts' needs and their expectations.

I agree to participate in this project, whose conditions are as follows:

- The project is aimed at making a tool for supporting energy experts find relevant housing cooperatives. For this purpose, semi-structured interviews will be conducted with Energy Experts.
- Interviews will last for about one hour and questions will deal with finding relevant housing cooperatives.
- The interview I give and the information it contains will be used solely for the purposes defined by the project.
- At any time, I can refuse to answer certain questions, discuss certain topics or even put an end to the interview without prejudice to myself.
- To facilitate the interviewer's job, the interview will be recorded. However, the recoding will be destroyed as soon as it has been transcribed.
- All interview data will be handled so as to protect their confidentiality. Therefore, no names will be mentioned and the information will be coded.
- All data will be destroyed at the end of the project.
- For any information about the project, I can contact [cristi@csc.kth.se].

Respondent's signature: _____ Date: _____

Interviewer's signature: _____ Date: _____

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